# Bernoulli equations

Bernoulli differential equations are another specific form of first order equations. To find the solution to a Bernoulli equation, we'll use a change of variables to convert the Bernoulli equation into a linear equation.

Once we've rewritten the equation in linear form, then of course we can find the solution to the linear equation. Finally, we'll make a simple substitution to put the solution equation back in terms of the variables we started with.

## Form of a Bernoulli differential equation

A Bernoulli differential equation is a differential equation in the form

$$y' + p(x)y = q(x)y^n$$

where $p(x)$ and $q(x)$ are continuous, and where $n$ is any real number. To solve Bernoulli equations, we'll always follow the same step-by-step process, which will be to use a change of variables in order to turn the Bernoulli equation into a linear equation.

If $n = 0$, then the Bernoulli equation is already a first order linear equation.

$$y' + p(x)y = q(x)y^0$$

$$y' + p(x)y = q(x)$$

If $n = 1$, then the Bernoulli equation is already a first order separable differential equation.

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$$y' + p(x)y = q(x)y^1$$

$$y' + p(x)y - q(x)y = 0$$

$$y' + (p(x) - q(x))y = 0$$

$$\frac{dy}{dx} + (p(x) - q(x))y = 0$$

$$\frac{dy}{dx} = -(p(x) - q(x))y$$

$$dy = -(p(x) - q(x))y\ dx$$

$$\frac{1}{y}\ dy = -(p(x) - q(x))\ dx$$

$$\frac{1}{y}\ dy = (q(x) - p(x))\ dx$$

In both of these cases, we won't have to do the extra work to convert the Bernoulli equation. But for any value $n \neq 0, 1$, our goal will be to perform a change of variables. In general, these are the steps we'll follow to find the solution:

1. If the Bernoulli equation isn't already given in standard form, rewrite it in standard form.
2. Divide both sides of the equation by $y^n$.
3. Identify the substitution $v = y^{1-n}$, implicitly differentiate to find $v' = (1 - n)y^{-n}y'$, and substitute into the Bernoulli equation.

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4. Put the resulting equation into the standard form of a linear differential equation.

5. Find the solution to the linear differential equation, then back-substitute for $v$ and solve for $y$.

Let's work through an example so that we can see how to find the solution to a Bernoulli differential equation.

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**Example**

Find the solution to the Bernoulli differential equation.

$$(y^4 + x^2y)\,dx - 6x^3\,dy = 0$$

If the equation isn't given in the standard form of a Bernoulli equation, our first step is always to rewrite it in standard form.

$$(y^4 + x^2y)\,dx = 6x^3\,dy$$

$$y^4 + x^2y = 6x^3\,\frac{dy}{dx}$$

$$\frac{dy}{dx} = \frac{y^4 + x^2y}{6x^3}$$

Split the fraction so that we can separate terms.

$$\frac{dy}{dx} = \frac{y^4}{6x^3} + \frac{x^2y}{6x^3}$$

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$$\frac{dy}{dx} - \frac{x^2 y}{6x^3} = \frac{y^4}{6x^3}$$

$$\frac{dy}{dx} - \frac{x^2}{6x^3} y = \frac{1}{6x^3} y^4$$

$$y' - \frac{1}{6x} y = \frac{1}{6x^3} y^4$$

With the equation in standard form, divide through by $y^n$. In this equation, that means we're dividing by $y^4$.

$$\frac{y'}{y^4} - \frac{1}{6x}\frac{y}{y^4} = \frac{1}{6x^3}\frac{y^4}{y^4}$$

$$y^{-4}y' - \frac{1}{6x} y^{-3} = \frac{1}{6x^3}$$

Our substitution is $v = y^{-3}$, so we'll differentiate to get

$$v' = -3y^{-4}y'$$

and then solve this for $y^{-4}y'$.

$$y^{-4}y' = -\frac{1}{3} v'$$

Now we can make substitutions into the Bernoulli equation.

$$y^{-4}y' - \frac{1}{6x} y^{-3} = \frac{1}{6x^3}$$

$$-\frac{1}{3} v' - \frac{1}{6x} v = \frac{1}{6x^3}$$

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Multiplying through by $-3$ puts the equation into standard form of a linear differential equation.

$$v' + \frac{1}{2x}v = -\frac{1}{2x^3}$$

To find the solution to the linear equation, we'll find the integrating factor,

$$I(x) = e^{\int p(x)\,dx}$$

$$I(x) = e^{\int \frac{1}{2x}\,dx}$$

$$I(x) = e^{\frac{1}{2}\ln x}$$

$$I(x) = e^{\ln(x^{\frac{1}{2}})}$$

$$I(x) = x^{\frac{1}{2}}$$

and then multiply through the linear equation by $I(x)$.

$$v' + \frac{1}{2x}v = -\frac{1}{2x^3}$$

$$v'x^{\frac{1}{2}} + \frac{1}{2x}x^{\frac{1}{2}}v = -\frac{1}{2x^3}x^{\frac{1}{2}}$$

$$v'x^{\frac{1}{2}} + \frac{1}{2}x^{-\frac{1}{2}}v = -\frac{1}{2}x^{-\frac{5}{2}}$$

Reverse the product rule for derivatives to rewrite the left side,

$$\frac{d}{dx}(vx^{\frac{1}{2}}) = -\frac{1}{2}x^{-\frac{5}{2}}$$

then integrate both sides.

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$$\int \frac{d}{dx}(vx^{\frac{1}{2}})\,dx = \int -\frac{1}{2}x^{-\frac{5}{2}}\,dx$$

$$vx^{\frac{1}{2}} = -\frac{1}{2}\left(-\frac{2}{3}\right)x^{-\frac{3}{2}} + C$$

$$vx^{\frac{1}{2}} = \frac{1}{3}x^{-\frac{3}{2}} + C$$

Solve for $v$.

$$v = \frac{1}{3}\left(\frac{x^{-\frac{3}{2}}}{x^{\frac{1}{2}}}\right) + \frac{C}{x^{\frac{1}{2}}}$$

$$v = \frac{1}{3x^2} + \frac{C}{x^{\frac{1}{2}}}$$

Find a common denominator in order to combine fractions.

$$v = \frac{1}{3x^2} + \frac{3Cx^{\frac{3}{2}}}{3x^2}$$

$$v = \frac{1 + 3Cx^{\frac{3}{2}}}{3x^2}$$

If $3C$ is some arbitrary constant, then we can simply use $C$ instead to represent that arbitrary constant.

$$v = \frac{1 + Cx^{\frac{3}{2}}}{3x^2}$$

Use $v = y^{-3}$ to back-substitute for $v$,

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$$y^{-3} = \frac{1 + Cx^{\frac{3}{2}}}{3x^2}$$

then solve for $y$.

$$\frac{1}{y^3} = \frac{1 + Cx^{\frac{3}{2}}}{3x^2}$$

$$y^3 = \frac{3x^2}{1 + Cx^{\frac{3}{2}}}$$

$$y = \sqrt[3]{\frac{3x^2}{1 + Cx^{\frac{3}{2}}}}$$

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